

3-MONTH DATA ANALYTICS ROADMAP

Excel + SQL + Python + Power BI - Build Real Projects Every Month

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3	12	3-4 hrs	1
Months	Weeks	Daily	Project/Month

DAILY STUDY PLAN (3-4 hours)

1 hr	1.5 hrs	1 hr
Theory + Learning	Practice + Coding	Project Build

GOLDEN RULES - FOLLOW THIS

X	Don't just watch tutorials -- open Excel/SQL/Python and type everything yourself
X	Don't skip SQL -- it is the most used skill in every data analytics job
OK	Work on REAL datasets -- download from Kaggle, not toy classroom examples
OK	Build 1 project every month -- put it on GitHub and LinkedIn immediately

TECH STACK + 3-MONTH OVERVIEW

M1	M2	M3
Excel + SQL Data Basics	Python + Pandas EDA + Viz	Power BI Dashboards + Jobs

MONTH 1 - Data Basics + Excel + SQL

Month 1	Data Basics + Excel + SQL
Week 1	■ What is Data Analytics -- descriptive, diagnostic, predictive analytics ■ Data lifecycle -- collection, cleaning, analysis, visualisation, decisions ■ Explore CSV datasets -- understand columns, data types, patterns ■ Goal: Understand what data means and how it is actually used ■ <i>...and more</i>
Week 2	■ Excel Basics -- SUM, AVERAGE, COUNT, IF, VLOOKUP, HLOOKUP ■ Data Cleaning in Excel -- remove duplicates, text functions, formatting ■ Advanced Excel -- Pivot Tables, Charts, conditional formatting ■ Practice: Sales dataset analysis end to end in Excel ■ <i>...and more</i>
Week 3	■ SQL Basics -- SELECT, WHERE, ORDER BY, GROUP BY, HAVING ■ Joins -- INNER JOIN, LEFT JOIN (most important for real work) ■ Aggregate functions -- COUNT, SUM, AVG, MIN, MAX with GROUP BY ■ Practice: Write queries on sample sales + customer databases ■ <i>...and more</i>
Week 4	■ SQL Advanced -- subqueries, nested queries, CASE statements ■ Window functions -- ROW_NUMBER, RANK, LAG, LEAD (basic intro) ■ Mini Project: Sales analysis + Customer analysis using SQL only ■ Goal: Solve real-world business queries with confidence ■ <i>...and more</i>
BUILD	■ Excel Sales Dashboard ■ SQL Sales + Customer Analysis ■ <i>...and more mini tasks</i>

MONTH 2 - Python + Data Analysis (VERY IMPORTANT)

Month 2	Python + Pandas + Data Analysis + Visualisation
Week 1	■ Python Basics -- variables, loops, functions, lists, dictionaries ■ Python for Data -- list comprehensions, file reading, error handling ■ Practice on HackerRank / Kaggle Python track -- do 20+ exercises ■ Resource: CodeWithHarry Python + freeCodeCamp Python full course ■ ...and more
Week 2	■ NumPy -- arrays, broadcasting, vectorised operations (no slow for-loops) ■ Pandas -- DataFrames, filtering, groupby, merge, handling missing data ■ Data Cleaning tasks -- duplicates, nulls, type conversion, string ops ■ Practice: Clean 2 messy real-world datasets from Kaggle ■ ...and more
Week 3	■ EDA (Exploratory Data Analysis) -- the most important analyst skill ■ Summary statistics -- mean, median, std dev, skewness, correlation ■ Outlier detection -- IQR method, z-score, visualising distributions ■ Practice: Analyse real datasets and write observations as comments ■ ...and more
Week 4	■ Matplotlib -- line chart, bar chart, histogram, scatter plot from scratch ■ Seaborn -- heatmap, pairplot, boxplot, beautiful statistical charts ■ Full Data Analysis Report -- clean data + EDA + charts + summary ■ Resource: Krish Naik (Matplotlib + Seaborn) + Corey Schafer ■ ...and more
BUILD	■ Full EDA Report on Real Dataset ■ Data Cleaning + Visualisation Project ■ ...and more

MONTH 3 - Advanced Analytics + Dashboards + Job Ready

Month 3	Advanced Analytics + Power BI / Tableau + Final Projects
Week 1	■ Trend Analysis -- moving averages, rolling windows, growth rates ■ Time Series Basics -- date handling in Pandas, resampling, plotting ■ Cohort Analysis + Funnel Analysis -- common in business analytics roles ■ Practice: Analyse 6-month sales data with trends and seasonality ■ ...and more
Week 2	■ KPIs and Business Metrics -- what analysts actually track at companies ■ Data Storytelling -- structure your findings like a business presentation ■ A/B Testing basics -- hypothesis testing, p-value concept for analysts ■ Goal: Convert raw numbers into clear business insights and decisions ■ ...and more
Week 3	■ Power BI / Tableau -- connect data, build visuals, create relationships ■ Dashboard Design -- filters, slicers, drill-downs, KPI cards ■ Build: Sales Dashboard + Customer Behaviour Dashboard in Power BI ■ Resource: CodeWithHarry (Power BI) + Alex The Analyst (best channel) ■ ...and more
Week 4	■ Final Project 1 -- Sales Analysis Dashboard (SQL + Python + Power BI) ■ Final Project 2 -- Customer Segmentation (EDA + clustering + charts) ■ Final Project 3 -- Business Insights Report with storytelling ■ Upload all 3 to GitHub -- write README, add screenshots, go job hunting ■ ...and more
BUILD	■ Sales Analysis Dashboard ■ Customer Segmentation Report ■ Business Insights Presentation

TARGET BREAKDOWN - What You Will Know After 3 Months

Phase	What You Will Know
Month 1	Excel proficiency + SQL queries for real databases + data intuition
Month 2	Python + Pandas data cleaning + full EDA + charts with Matplotlib/Seaborn
Month 3	Power BI dashboards + business thinking + 3 portfolio projects + job-ready

RESOURCES - Everything You Need to Become a Data Analyst

PRACTICE PLATFORMS

Kaggle	kaggle.com	Best platform for real datasets, notebooks, and competitions -- use for every project
HackerRank SQL	hackerrank.com/domains/sql	Structured SQL practice problems -- complete all Easy and Medium before interviews
LeetCode SQL	leetcode.com/problemset/database	Interview-level SQL questions -- essential for data analyst job prep at top companies
Syntax Error	syntaxerrorr.com	Our website -- notes, one-shot videos, placement resources for analytics + DSA

YOUTUBE RESOURCES (Topic-wise)

EXCEL (DETAILED)	ExcelsFun by Mike Girvin youtube.com/@excelisfun The most detailed Excel channel on YouTube. Every function, formula and Pivot Table explained step by step. Use for Week 2 of Month 1 -- extremely thorough.
EXCEL + POWER BI	Leila Gharani by Leila Gharani youtube.com/@LeilaGharani Best channel for advanced Excel, Power Query and Power BI. Clean professional teaching style. Go here after ExcelsFun when you need dashboards and automation.
SQL - HINDI	Apna College SQL by Apna College youtube.com/@ApnaCollegeOfficial Hindi SQL course -- beginner friendly, structured from basics to advanced. Best for Indian learners starting SQL in Week 3. Clear examples, interview-focused content.
SQL + PYTHON	freeCodeCamp SQL by freeCodeCamp.org youtube.com/@freecodecamp Free full-length SQL + Python courses -- covers everything from SELECT to window functions. Watch alongside Apna College for English explanations and deeper concepts.
PYTHON + POWER BI	CodeWithHarry by Code With Harry youtube.com/@CodeWithHarry Hindi Python + Power BI tutorials -- structured series from basics to full data analysis. Perfect for Month 2 Python and Month 3 Power BI in one familiar channel.
FULL ANALYST PATH	Alex The Analyst by Alex Freberg youtube.com/@AlexTheAnalyst The #1 channel for data analyst career prep -- SQL, Python, Power BI, Tableau, portfolio projects and how to land your first job. Watch the full analyst bootcamp series.
EDA + VISUALISATION	Krish Naik by Krish Naik youtube.com/@krishnaik06 Best for Matplotlib, Seaborn and EDA tutorials. Very practical approach with real datasets. Use for Month 2 Weeks 3-4 when you are building visualisation skills.

MATPLOTLIB +
SEABORN

Corey Schafer by Corey Schafer

youtube.com/@coreyms

Clean and professional Python + Matplotlib/Seaborn tutorials. Great for understanding the WHY behind the code. Use alongside Krish Naik for data visualisation.

SYNTAX ERROR

Our Website

syntaxerrorr.com

Our Instagram

[@code.abhii07](https://www.instagram.com/code.abhii07)

*"Data is the new oil. But raw data is crude -- YOU are the refinery. Open your laptop and start analyzing.
The insights are waiting."*